Next Bill Amount Prediction

From raw billing history to a model we can trust, and where it still falls short.

The tuned model misses a typical bill by **163.5**, against **179.1** for the best simple rule. That is **8.7**% better, and it wins in **7** of **7** unseen months. It cannot see sudden spikes coming.

THE QUESTION

Can we predict next month's bill before it is issued?

Each customer is billed every month. The goal is to predict the next bill from the customer's plan, usage and bill history, early enough to act on it.

Customers
1,222

Bills in the data
2,000

Months covered
32

Months held out for testing
7
Never used for any choice

The dataset is synthetic, so the results show the method, not a live business.

THE DATA

Plans set the level. Spikes make it hard.

A bill follows the plan a customer is on, then drifts with add-ons and usage. A few bills run far above normal and pull the average up.

Median bill
708.0

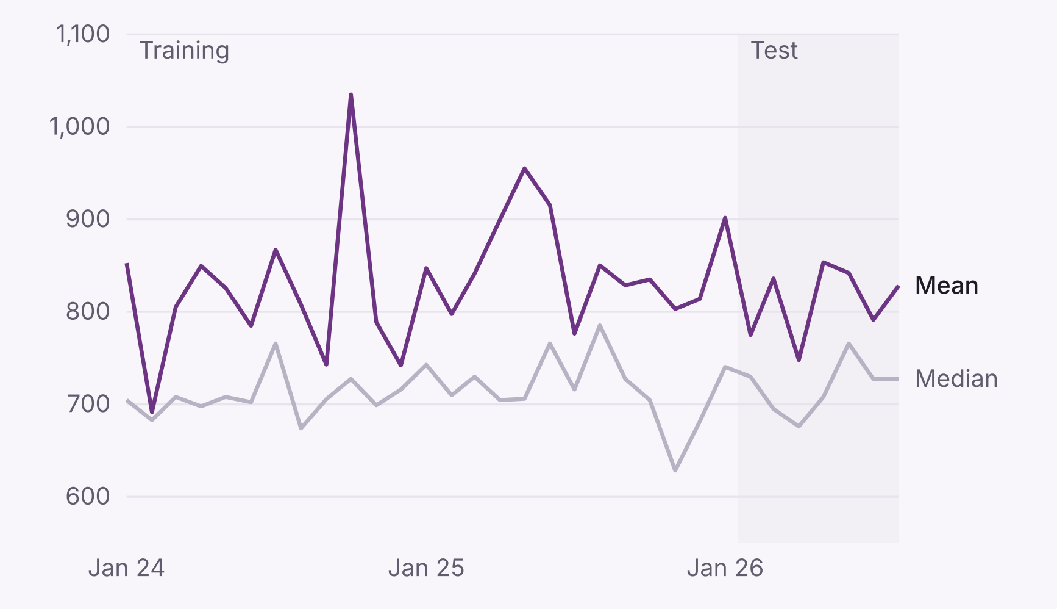
Mean bill
829.3
Pulled up by spikes

Largest bill
5,284.6

Repeat a recent bill
45.1%
Next bill equals one of the last three

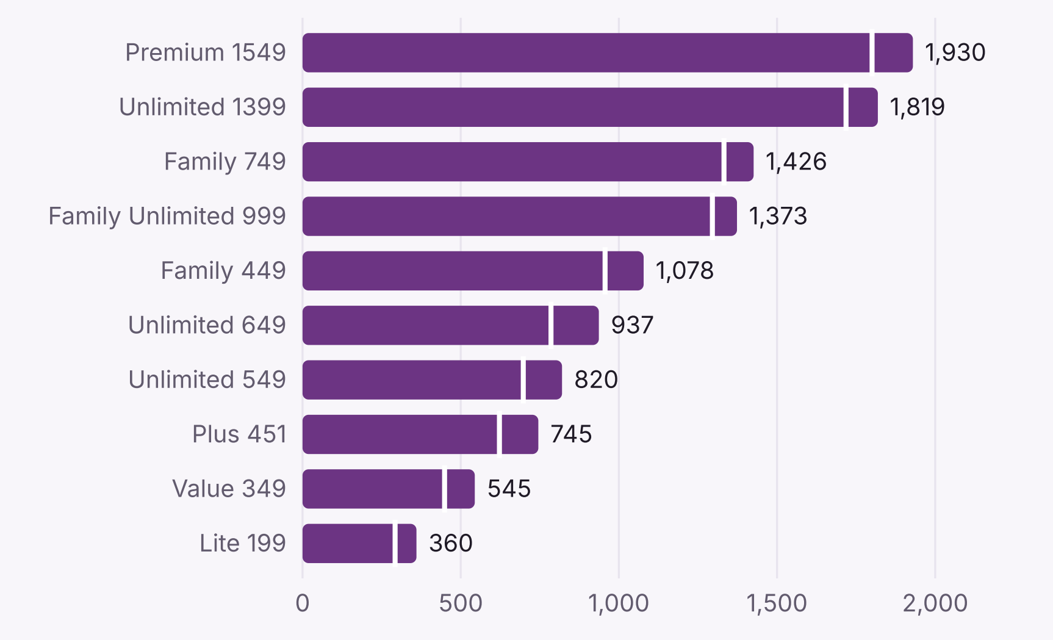
Average bill by month

Flat on the whole, with a few jumps. The shaded months are the test period.



Average bill by plan

Bar is the mean, tick is the median. Plan sets the level.



THE APPROACH

Every choice guards against fooling ourselves.

For each step: what we chose, why, and the tempting option we rejected.

Split by time

WHY

Real forecasts only know the past. Early months train the model; the last months are the final exam.

NOT THIS

A random split. Later bills would help predict earlier ones and flatter the score.

Judge by MAE

WHY

It is the typical miss in bill units, and a few huge spikes cannot dominate it.

NOT THIS

RMSE or R^2 alone. Squaring errors lets rare spikes pick the winner. Both are still reported.

Validate on rolling months

WHY

Each fold trains on the past and tests on the next few months, as it would in production.

NOT THIS

Shuffled k-fold. It mixes future and past and hides the drift.

Prepare data on training only

WHY

Clipping limits, plan encoding and scaling are learned from the training months, then applied to the test months.

NOT THIS

Fitting on all rows first. That quietly lets the test months shape the model.

Beat simple rules first

WHY

The simple rules set the bar. A model that cannot beat the typical bill for a plan is not worth shipping.

NOT THIS

Comparing models only with each other.

Keep only features that help

WHY

Engineered features cut the error by only **0.03**%, feature selection by **0.14**%, and PCA with **27** components was worse. None was adopted.

NOT THIS

Adding features because they sound clever.

Rolling validation folds

Every fold trains on all earlier months and validates on the next block. The final test months sit after all of them.

Fold	Training bills	Validation bills	Trained until	Validated on
1	557	249	Sep 2024	Oct 2024 - Jan 2025
2	806	188	Jan 2025	Feb - Apr 2025
3	994	189	Apr 2025	May - Jul 2025
4	1,183	190	Jul 2025	Aug - Oct 2025
5	1,373	190	Oct 2025	Nov 2025 - Jan 2026

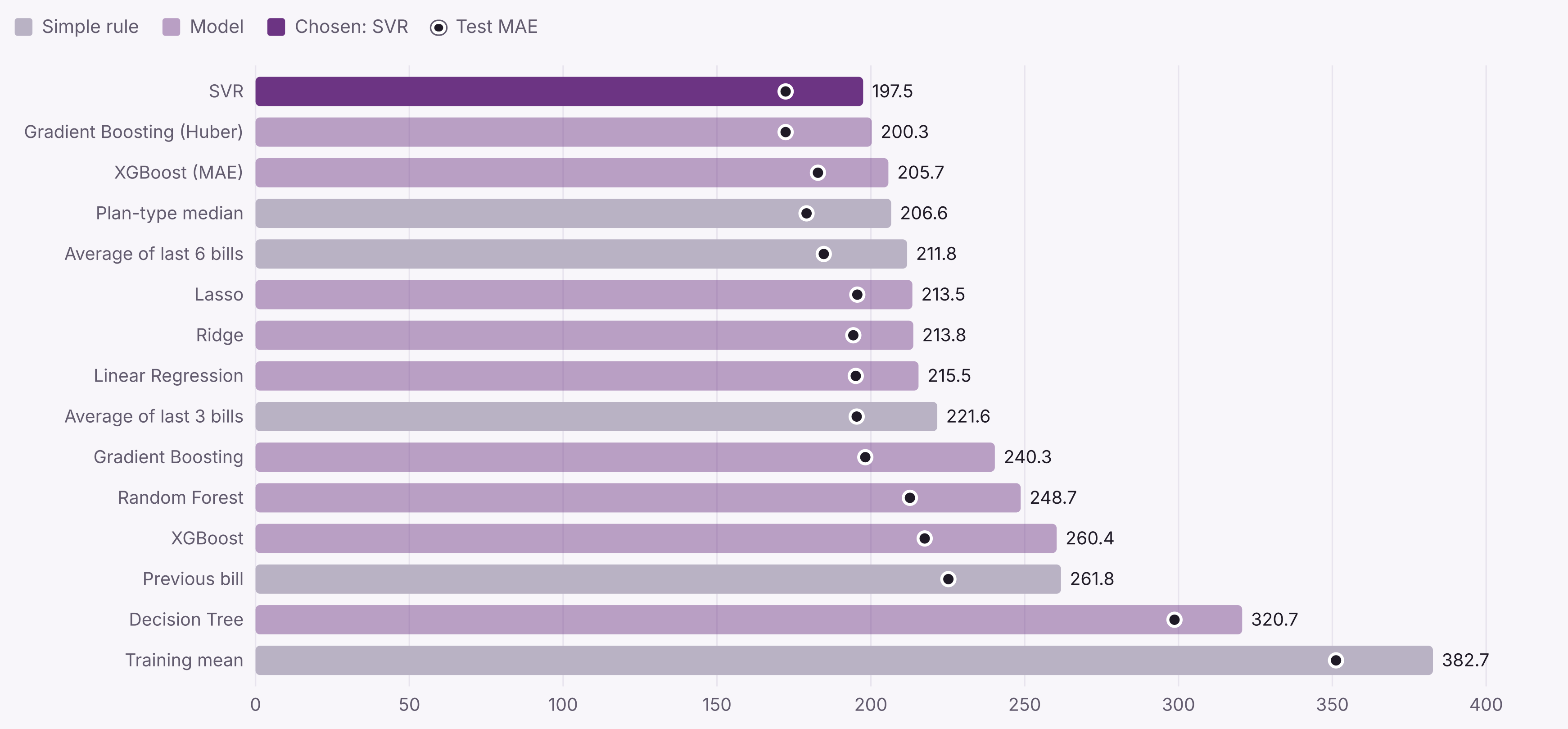
THE MODELS

Only robust models beat the simple rule. Tuning widened the gap.

The models were ranked on rolling validation MAE. The models with a gentle loss lead. The same boosting methods with squared error fall far behind.

Validation MAE, lower is better

Bars are validation MAE, the score we ranked on. Dots are test MAE, looked at only afterwards.



Tuning the four best

Search ran on the same rolling folds. SVR stayed first after tuning.

Model	Validation, default	Validation, tuned	Test, default	Test, tuned
SVR	**197.5**	**184.5**	**172.3**	**163.5**
XGBoost (MAE)	205.7	187.0	182.8	167.6
Gradient Boosting (Huber)	200.3	190.0	172.3	168.8
Lasso	213.5	212.5	195.6	194.0

Why SVR

SVR ignores very small errors and charges only linearly for large ones, so rare spikes do not drag the fit. It also won validation, the score we ranked on. Tuned settings: C **1,000**, epsilon **10**, gamma **0.003**.

Linear models score higher on test R^2 and lower on RMSE (Ridge: **0.546** and **315.8**, against **0.508** and **328.7** for the tuned SVR). We ranked by validation MAE because it reflects the typical bill.

THE RESULT

On unseen months the model beats the best rule, every month.

Scores on the held-out test months, next to the best simple rule: the median bill for the plan.

Typical miss (MAE)
163.5
Best rule: **179.1**

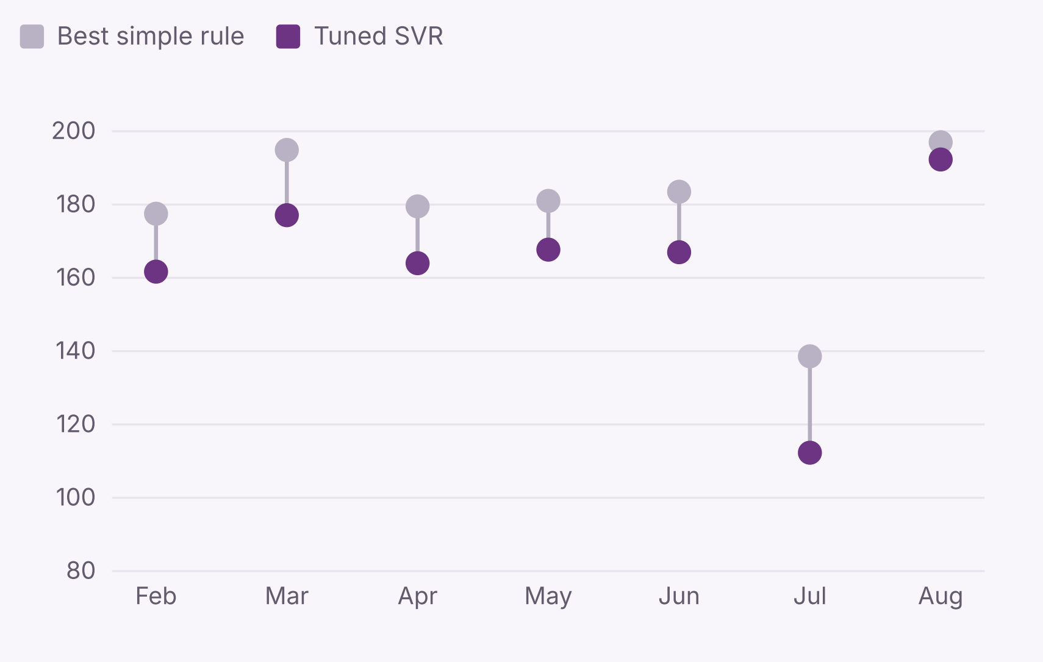
RMSE
328.7
Best rule: **335.0**

R^2
0.508
Best rule: **0.489**

Error as share of bills
20.2%
Best rule: **22.1**%

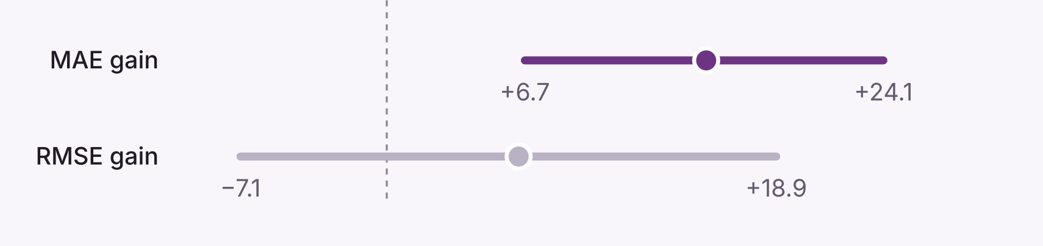
Error by test month

MAE per month. The model is lower than the rule in every month.



How sure is the gain?

Improvement over the rule, with a confidence range from resampling the test bills. A range that stays right of zero is a real gain.



The MAE gain is reliable. The RMSE gain is not clear, because its range crosses zero. Training MAE is **157.0** against **163.5** on test, so there is no sign of overfitting.

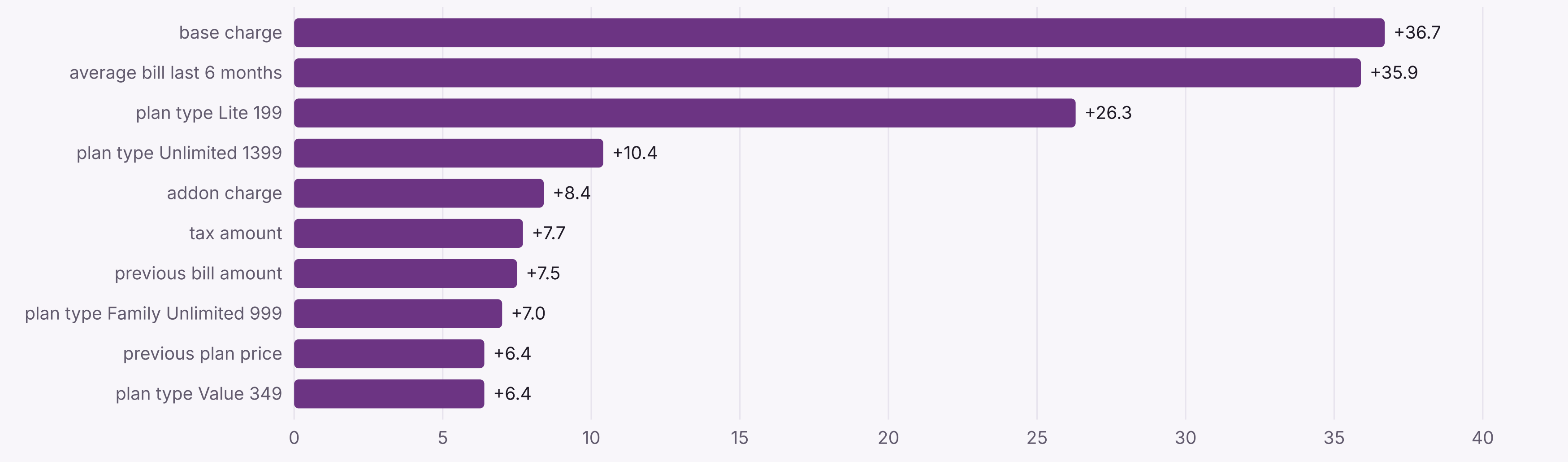
THE DRIVERS

Base charge, recent average and the cheapest plan matter most.

Permutation importance: how much the error grows when one feature is scrambled. The bill follows the plan level and the customer's own recent history.

Most important features

Increase in test MAE when the feature is shuffled.



THE LIMITS

Sudden spikes stay out of reach.

The model is steady on ordinary bills. It under-predicts the rare bill that jumps far above the customer's history.

Share of error from the worst tenth of bills
53.6%

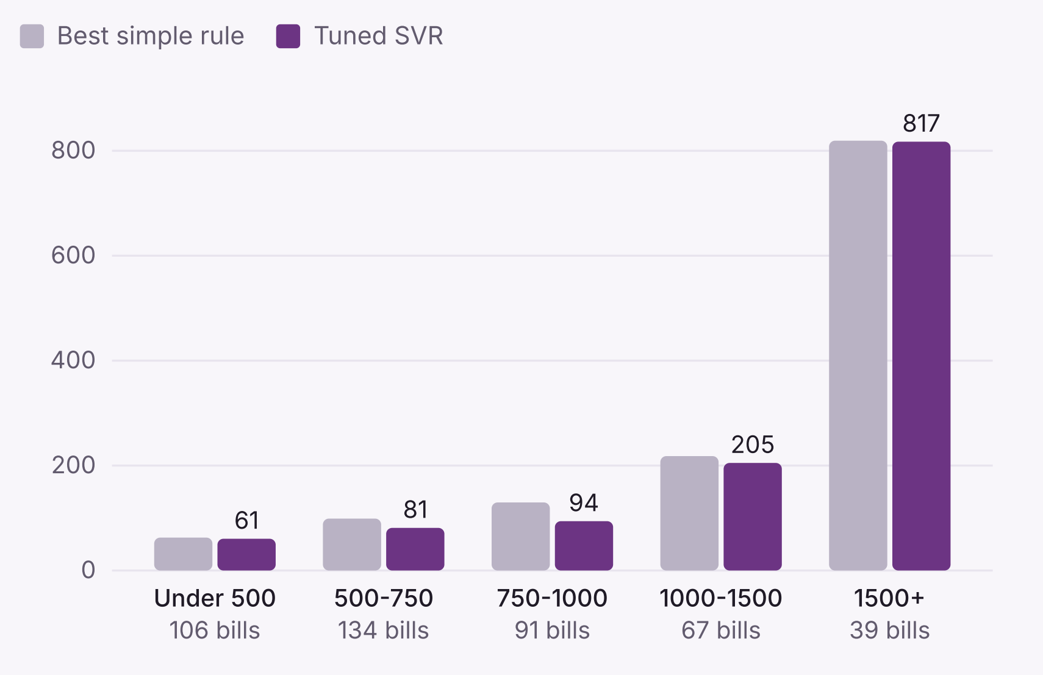
Test bills that are high bills
11.2%
At least one and a half times their own recent average

Interval coverage
75.5%
Target **80**%

Average interval width
506.9

Error by actual bill size

MAE grows fast for the largest bills, and the model barely beats the rule there.



Ranking likely high bills

A watch list ranks customers by predicted bill against their own average. It beats chance by a small margin.

Ranking score	ROC-AUC	Hits in top 25	Hits in top 100
Model: predicted bill / 6-month average	0.565	0.20	0.19
Naive: previous bill / 6-month average	0.540	0.12	0.11
Random ranking	0.500	0.11	0.11

With the chosen threshold the flag marked **51** customers, with precision **0.176** and recall **0.184**. Treat it as a watch list, not an alarm.

The prediction ranges aim to cover **80**% of test bills but cover **75.5**%, so they are a little too narrow. Use them as a rough guide to spread.

TAKEAWAY

Use it for typical bills. Plan for spikes separately.

Use it for

- Estimating the next bill for budgeting
- Showing a likely range, not a single figure
- Beating the plan-median rule on ordinary bills

Do not rely on it for

- Warning about sudden large bills
- Picking out the single customer who will spike
- Claims about a live business, since the data is synthetic

Next steps

- Add signals for one-off charges and plan changes
- Retrain each month on the rolling window
- Re-check interval coverage and widen if it stays low

Figures come from the Next Bill Amount Prediction notebook. Test results use the held-out final months only, validation results use rolling folds on the earlier months.